



CalPoly
Pomona

**WISTLS SAMPLING DRILL
DESIGN AND DEVELOPMENT**

By
Luke Esperiquette

Faculty Advisor:
Professor Behnam Bahr

Department of Mechanical Engineering
California State Polytechnic University, Pomona

Submitted on December 11, 2025

TABLE OF CONTENTS

Acknowledgements.....	iii
Symbols and Abbreviations	iii
Abstract.....	v
1 Introduction.....	1
2 Design and construction/simulation.....	4
3 Testing and results	11
4 Conclusions.....	11
5 Division of Responsibilities	21
References.....	24
Appendix A.....	25

ACKNOWLEDGEMENTS

I would like to thank the NASA Jet Propulsion Laboratory (JPL) for providing the project scope and technical requirements for the WISTLS (Water Isotope Tunable Laser Spectrometer) sampling drill. I am especially grateful to Kalind Carpenter, Lance Christensen, and Kevin Schwarm for their guidance on lunar regolith mechanics, drilling physics, and instrument integration. I would like to thank Professor Behnam Bahr, my Faculty Advisor, for his support of this project. Finally, I would like to acknowledge the Mechanical Engineering Department at Cal Poly Pomona for providing the resources and facilities that made this work possible.

SYMBOLS AND ABBREVIATIONS

Symbols

D : Diameter (mm)

FOS : Factor of Safety (unitless)

J : Polar Moment of Inertia (m^4)

r : Radius (m)

S : Strength (MPa)

T : Torque ($N \cdot m$)

V : Volume (cm^3)

ρ : Density ($\frac{g}{cm^3}$)

τ : Shear Stress or Shear Strength (MPa)

ω : Angular Velocity ($\frac{rad}{s}$)

Abbreviations

CAD: Computer-Aided Design

FOS: Factor of Safety

JPL: Jet Propulsion Laboratory

OD: Outer Diameter

ISRU: In-situ Resource Utilization

WISTLS: Water Isotope Tunable Laser Spectrometer

ABSTRACT

This project presents the design and development of a compact auger drill for collecting regolith samples on the lunar surface as part of the WISTLS mission. The drill must collect at least 20 grams of material while operating under severe constraints: cryogenic temperatures, minimal power, and the tight mass and volume limits of small robotic landers.

The design uses a 30 mm outer diameter configuration that balances sample capacity, structural strength, and integration with the WISTLS instrument. The cutting bit features dual 100° lobes designed to penetrate compacted lunar soil with minimal torque. Helical auger flights transport material upward along the shaft, with the pitch and geometry optimized for sample retention in the low-gravity lunar environment. The design is tailored for metal additive manufacturing, accounting for support structures, minimum feature sizes, and thermal management during fabrication.

Early structural analysis revealed a critical flaw: the original Aluminum 6061-T6 drive shaft could not handle the expected loads. Under an estimated peak torque of 5 N·m, it yielded a Factor of Safety of only 0.88, unacceptable for mission reliability. The shaft was redesigned using Titanium alloy (Ti-6Al-4V) with an increased diameter of 5 mm, raising the Factor of Safety to 2.70. This change maintained compatibility with other components while providing the necessary strength-to-weight ratio for a payload-constrained lunar mission.

The final design includes a 30 mm cutting bit, helical auger flights, a 5 mm Ti-6Al-4V drive shaft, and a protective sheath to prevent sample loss. Torsional stress analysis confirmed the titanium shaft experiences a maximum shear stress of 203 MPa under design loads, well below the yield strength. Volumetric analysis verified the auger can exceed the 20 gram requirement.

The design is manufacturing-ready and satisfies all mechanical, operational, and manufacturing requirements for lunar regolith sampling.

1 INTRODUCTION

Future lunar missions are increasingly focused on In-Situ Resource Utilization (ISRU) as a way to reduce mission costs, extend operational timelines, and minimize dependence on Earth-supplied resources. A key scientific goal is characterizing subsurface water and volatile distribution in the lunar regolith. Understanding these distributions provides insight into the Moon's thermal and geological history while informing the requirements for sustained human presence [10]. The WISTLS (Water Isotope Tunable Laser Spectrometer) instrument was developed to address this need, requiring a compact, low-power drill that can retrieve subsurface samples without thermally altering volatile species. Designing such a mechanism within the tight volume, mass, and power constraints of small robotic landers presents a significant mechanical engineering challenge.

Lunar regolith behaves very differently from terrestrial soils. It is highly compacted, composed of angular particles formed by micrometeorite impacts, and is extremely abrasive during cutting operations. Documented shear strengths range from 1 to 5 MPa depending on depth and compaction, with internal friction angles often exceeding 35° [1,2]. These properties directly affect the cutting forces during drilling and determine the mechanical loads on structural components. Cutting torque, T , can be approximated using simplified soil-tool interaction models of the form

$$T \propto \tau \left(\frac{\pi D^2}{4} \right) L \quad (1)$$

where τ is regolith shear strength, D is bit diameter, and L is the effective engagement length of the cutting surface [9]. While this equation doesn't capture all geometric effects like tip angle or flute geometry, it establishes the primary relationships between material properties, drill scale, and mechanical loading. These relationships guided the initial sizing decisions during design.

To evaluate the structural adequacy of candidate shaft geometries, torsional shear stress can be calculated using

$$\tau = \frac{Tr}{J} \quad (1)$$

$$J = \frac{\pi r^4}{2} \quad (3)$$

where J is the polar moment of inertia for a solid circular cross-section [3,8]. This relationship was applied iteratively to evaluate different materials and dimensions. Together, these equations provide the theoretical framework linking the regolith environment, drill geometry, and material stress response.

The objective of this project is to design a compact regolith sampling drill that meets the mechanical and operational requirements established by NASA's Jet Propulsion Laboratory for the WISTLS instrument. The drill must collect at least 20 grams of regolith, operate at minimal torque to reduce power consumption, withstand the abrasive lunar environment, and be manufacturable through metal additive processes. The design process involved iterative geometry development, material selection analysis, and analytical validation to arrive at a configuration that satisfies all requirements. Beyond meeting mission-specific criteria, this work contributes to the broader development of cost-effective lunar exploration technologies and supports ISRU strategies for future crewed missions.

The design methodology draws on established principles from mechanics of materials [3], granular mechanics [2,9], machine component design [4], and additive manufacturing considerations. Material properties for candidate alloys were obtained from industry standards and mechanical testing protocols [5-7]. The analysis integrates empirical data from NASA's Handbook of Lunar Materials [1], literature on regolith drilling and granular flow [2,9], and foundational references on torsional mechanics [3,8]. These sources informed geometric design decisions, guided material selection, and provided the analytical basis for validating that the final configuration is both technically sound and aligned with current lunar science objectives.

2 DESIGN AND CONSTRUCTION/SIMULATION

This section describes the iterative design process and constraint-driven decisions that produced the final WISTLS Sampling Drill configuration. The design prioritizes structural reliability, sample collection capacity, and manufacturing feasibility within the strict geometric and mass limitations of small robotic lander platforms.

2.1 DESIGN SOLUTION AND REQUIREMENTS FULFILLMENT

The WISTLS Sampling Drill uses an auger-based core-and-sheath architecture designed to meet NASA's volumetric and sample integrity requirements for lunar volatile collection. The design addresses three primary objectives established during the requirements phase.

Geometric constraint compliance. The outer diameter was set at 30 mm to balance sample collection capacity with structural requirements while maintaining a compact form factor. The final design uses a 30 mm diameter cutting bit with a 29 mm outer sheath, maximizing the sampling volume while maintaining clearance for rotation and thermal expansion. This dimension was selected to provide adequate space for the required sample mass while remaining compatible with small robotic lander platforms.

Sample mass requirement. A minimum of 20 grams is required for isotopic analysis, corresponding to approximately 13.3 cm³ based on lunar regolith density of 1.5 g/cm³ [1]. The effective sampling length was set at 100 mm, yielding a theoretical collection volume of 15.1 cm³. This provides a 13% margin above the minimum to account for packing efficiency and material loss during extraction (detailed volumetric analysis is in Appendix A).

Structural integrity and manufacturing optimization. The center drive shaft is the most structurally critical component, as it transmits the full cutting torque under torsional loading. Titanium alloy (Ti-6Al-4V) was selected to achieve a Factor of Safety of 2.70 against shear failure, as validated in Section 3. To maintain structural continuity at the highly stressed shaft-to-bit interface, the drill bit body (Item 3 in Table 2) was also specified as titanium. The actual cutting is performed by two small carbide steel inserts (Item 3a) that mount to the titanium body via threaded fasteners. This modular approach limits the use of difficult-to-machine carbide to only the cutting edges, significantly reducing manufacturing cost and complexity while maintaining wear resistance where it matters most.

During the design process, non-functional prototypes of key components were fabricated using additive manufacturing to verify geometric compatibility, identify assembly issues, and confirm fit within the external integration envelope. These physical models enabled refinement of the auger geometry and validation of sheath clearances before finalizing dimensions for the flight-ready design. All structural analysis is based on the material properties and dimensions of the final specification in Table 2.

The geometric configuration from the completed design iteration is summarized in Table 1. These dimensions represent the optimized balance between sample capacity, structural performance, and manufacturing feasibility.

Table 1: Final Geometric Specifications for WISTLS Sampling Drill

Component	Parameter	Value	Unit
Sheath			
	Outer Diameter	29.0	mm
	Inner Diameter	21.0	mm
	Wall Thickness	4.0	mm
	Effective Length	100.0	mm
	Material	Aluminum 6061-T6	
Cutting Bit			
	Diameter	30.0	mm
	Angular Coverage	200	degrees
	Material	Titanium (Ti-6Al-4V)	
Center Drive Shaft			
	Diameter	5.0	mm
	Material	Titanium (Ti-6Al-4V)	
Auger Tube			
	Outer Diameter	7.5	mm
	Inner Diameter	5.5	mm
	Wall Thickness	1.0	mm
	Clearance to Center Shaft	0.25	mm
	Material	Aluminum 6061-T6	
Auger Fin			
	Outer Diameter	20.5	mm
	Radial Height	6.5	mm
	Helical Pitch	50.8	mm
	Angular Coverage	180	degrees
	Clearance to Sheath	0.25	mm
	Material	Aluminum 6061-T6	
Sample Collection Performance			
	Theoretical Volume	30.2	cm ³
	Effective Sample Volume	15.1	cm ³
	Estimated Sample Mass ($\rho = 1.5 \text{ g/cm}^3$)	22.7	g
	Margin Above Requirement (20 g)	13.5	%

2.2 DESIGN CONSTRAINTS AND EXTERNAL CONSIDERATIONS

Multiple engineering constraints and external factors influenced the final design decisions, requiring careful evaluation of competing priorities.

Engineering constraints. The 30 mm outer diameter was established during the initial design phase to balance sample collection capacity with mass and structural requirements. This dimension provides adequate volumetric capacity for the 20 gram sample requirement while maintaining a compact form factor suitable for integration with small robotic landers. Mass minimization was prioritized throughout the design to reduce launch costs and maintain compatibility with typical payload mass allocations for lunar missions. The effective sampling length of 100 mm was selected to achieve the required sample mass while limiting total assembly mass to 36.3 grams. Low-power consumption was addressed through optimization of auger geometry and selection of conservative rotational velocities to minimize cutting torque.

Public health, safety, and welfare considerations. In space exploration missions, these considerations relate primarily to mission reliability and planetary protection protocols. The high Factor of Safety ensures the drill will not fail catastrophically and create debris that could contaminate the lunar surface or interfere with other mission systems. Material selection based on flight-heritage standards (Ti-6Al-4V) ensures reliable operation in the extreme thermal and vacuum environment of the lunar surface.

Environmental and economic factors. The selection of titanium alloy represents a necessary economic trade-off, as the higher material and manufacturing costs are justified by the structural

requirements for mission success. The design supports broader economic objectives associated with in-situ resource utilization by demonstrating the technical feasibility of collecting lunar volatiles. Successful implementation of such technologies has substantial long-term economic impact by reducing the logistical burden and cost of transporting consumables from Earth to support sustained lunar operations.

Engineering standards and codes. Structural analysis adheres to standard machine design principles, with Factor of Safety calculated against shear yield strength as defined in mechanics of materials references [3]. Material properties were obtained from industry standards and handbook data [6,7] to ensure consistency with accepted engineering practice.

2.3 ANALYTICAL FRAMEWORK

Structural validation relies on calculating the maximum torsional shear stress in the 5 mm diameter center shaft underestimated cutting torque. The analytical model uses the following relationships.

Torsional shear stress at radius r due to applied torque T :

$$\tau = \frac{T \cdot r}{J} \quad (2)$$

Polar moment of inertia for a solid circular shaft of radius r :

$$J = \frac{\pi r^4}{2} \quad (3)$$

Factor of Safety based on material allowable stress:

$$\text{FOS} = \frac{S_{\text{allowable}}}{S_{\text{applied}}} \quad (4)$$

Cutting torque approximation based on regolith shear strength τ_{soil} , bit diameter D , engagement length L , and geometry factor K :

$$T \approx \tau_{\text{soil}} \cdot \frac{\pi D^2}{4} \cdot L \cdot K \quad (5)$$

These expressions establish the computational framework for evaluating shaft adequacy under operational loading, as detailed in Section 3.

2.4 BILL OF MATERIALS

The components, materials, and estimated masses for the mission-ready configuration are summarized in Table 2. The modular cutting tip design (Items 3 and 3a) reflects the manufacturing optimization strategy discussed in Section 2.1. Mass estimates incorporate the 100 mm effective length specification.

Table 2: Bill of Materials

Item	Part Name	Qty	Material	Manufacturing Process	Mass (g)
1	Outer Sheath Tube	1	Al 6061-T6	Additive Manufacturing	14.3
2	Auger Flights	1	Al 6061-T6	Additive Manufacturing	3.5
3	Drill Bit Body / Drive Shaft	1	Ti-6Al-4V	Additive Manufacturing	12
3a	Carbide Cutting Inserts	2	Carbide Steel	Machining	2.0
4	Motor Coupling Mount	1	Ti-6Al-4V	Additive Manufacturing	4.5
	Total Estimated Mass				36.3

3 TESTING AND RESULTS

This project relied on analytical validation rather than physical testing due to the unavailability of lunar regolith simulants and specialized drilling test equipment within the project timeline.

The structural performance of the drill was evaluated through hand calculations and engineering analysis based on established mechanics of materials principles. This section presents the structural validation results that confirmed the adequacy of the final design configuration.

3.1 STRUCTURAL VALIDATION METHODOLOGY

The primary structural concern for the drill assembly is torsional failure of the center drive shaft during cutting operations. To evaluate this risk, a systematic analysis was performed comparing two candidate materials: Aluminum 6061-T6 and Titanium Ti-6Al-4V. The analysis followed these steps:

1. Estimate the maximum cutting torque based on lunar regolith properties and bit geometry
2. Calculate the polar moment of inertia for the proposed shaft diameter
3. Determine the maximum shear stress at the shaft surface under applied torque
4. Calculate the Factor of Safety by comparing applied stress to material yield strength
5. Verify that the Factor of Safety meets or exceeds acceptable design criteria

3.2 TORQUE ESTIMATION

The cutting torque was estimated using the simplified soil-tool interaction model from Equation (4):

$$T \approx \tau_{\text{soil}} \cdot \frac{\pi D^2}{4} \cdot L \cdot K$$

Using conservative assumptions for lunar regolith with $\tau_{\text{soil}} = 3 \text{ MPa}$ (mid-range value from literature [1,2]), bit diameter $D = 30 \text{ mm}$, engagement length $L = 10 \text{ mm}$, and a geometry factor $K = 0.5$ to account for the dual-lobe cutting configuration, the estimated peak torque is approximately $5 \text{ N}\cdot\text{m}$. This represents a conservative upper bound for drilling in compacted regolith.

3.3 INITIAL DESIGN ANALYSIS – ALUMINUM SHAFT

The preliminary design specified an Aluminum 6061-T6 drive shaft with a diameter of 4 mm.

The structural adequacy was evaluated as follows:

Polar moment of inertia:

$$J = \frac{\pi r^4}{2} = \frac{\pi (0.002)^4}{2} = 2.51 \times 10^{-11} \text{ m}^4$$

Maximum shear stress at surface ($r = 0.002 \text{ m}$):

$$\tau = \frac{T \cdot r}{J} = \frac{5 \cdot 0.002}{2.51 \times 10^{-11}} = 398 \text{ MPa}$$

Material properties for Aluminum 6061-T6 [6]:

Shear yield strength: $S_{sy} \approx 0.577 \times \sigma_y = 0.577 \times 276 \text{ MPa} = 159 \text{ MPa}$

Factor of Safety:

$$\text{FOS} = \frac{159}{398} = 0.40$$

This result indicated severe inadequacy. A Factor of Safety below 1.0 means the shaft would yield under the estimated loading, presenting unacceptable risk of mechanical failure during mission operations.

3.4 DESIGN REVISION – TITANIUM SHAFT

To address the structural deficiency, two modifications were implemented:

1. Increase shaft diameter from 4 mm to 5 mm
2. Change material from Aluminum 6061-T6 to Titanium Ti-6Al-4V

The revised configuration was analyzed as follows:

Polar moment of inertia (revised):

$$J = \frac{\pi r^4}{2} = \frac{\pi (0.0025)^4}{2} = 6.14 \times 10^{-11} \text{ m}^4$$

Maximum shear stress at surface ($r = 0.0025 \text{ m}$):

$$\tau = \frac{T \cdot r}{J} = \frac{5 \cdot 0.0025}{6.14 \times 10^{-11}} = 203 \text{ MPa}$$

Material properties for Titanium Ti-6Al-4V [7]:

Shear yield strength: $S_{sy} \approx 0.577 \times \sigma_y = 0.577 \times 950 \text{ MPa} = 548 \text{ MPa}$

Factor of Safety (revised):

$$\text{FOS} = \frac{548}{203} = 2.70$$

The revised configuration achieves a Factor of Safety of 2.70, which provides adequate margin against structural failure while accounting for uncertainties in torque estimation and material properties.

3.5 RESULTS SUMMARY AND PERFORMANCE VALIDATION

Table 3 summarizes the comparative structural performance of the two design iterations. The results clearly demonstrate that the titanium shaft configuration meets the structural requirements for safe operation.

Table 3: Structural Analysis Results Comparison

Parameter	Aluminum 6061-T6 (Initial)	Titanium Ti-6Al-4V (Final)	Unit
Shaft Diameter	4.0	5.0	mm
Polar Moment of Inertia	2.51×10^{-11}	6.14×10^{-11}	m ⁴
Applied Torque	5.0	5.0	N·m
Maximum Shear Stress	398	203	MPa
Material Shear Yield Strength	159	548	MPa
Factor of Safety	0.40	2.70	-
Status	Inadequate	Acceptable	-

The analytical validation confirms that the final design configuration satisfies the structural requirements for lunar drilling operations. The Factor of Safety of 2.70 provides a reasonable margin against uncertainty in loading conditions while avoiding unnecessary over-design that would increase mass beyond mission constraints.

3.6 DESIGN MODIFICATIONS

Based on the structural analysis results, the following design modifications were implemented:

- 1. Shaft material changed from Aluminum 6061-T6 to Titanium Ti-6Al-4V** – This change increased the shear yield strength from 159 MPa to 548 MPa, providing substantially greater load-carrying capacity.
- 2. Shaft diameter increased from 4 mm to 5 mm** – This geometric modification increased the polar moment of inertia by 145%, reducing the applied shear stress for a given torque.
- 3. Drill bit body material aligned with shaft material** – To maintain structural continuity at the shaft-to-bit interface and avoid stress concentration at dissimilar material joints, the drill bit body was also specified as Titanium Ti-6Al-4V.

These modifications increased the total assembly mass by approximately 8 grams compared to the all-aluminum configuration, but this trade-off was deemed necessary to ensure mission reliability. The mass increase remains within acceptable limits for small robotic lander platforms.

4 CONCLUSIONS

This project successfully developed a compact auger drill design for lunar regolith sampling that meets the operational requirements established by NASA's Jet Propulsion Laboratory for the WISTLS instrument. The primary objective, to design a drill capable of collecting at least 20 grams of regolith while operating within strict mass, power, and geometric constraints, was achieved through iterative design and analytical validation.

The final configuration uses a 30 mm diameter cutting bit with helical auger flights and a 5 mm diameter Titanium Ti-6Al-4V drive shaft. Structural analysis confirmed that the titanium shaft achieves a Factor of Safety of 2.70 under the estimated peak torque of 5 N·m, providing adequate margin against mechanical failure during drilling operations. Volumetric analysis verified that the auger geometry can collect approximately 22.7 grams of material over a 100 mm sampling depth, exceeding the minimum requirement by 13.5%. The total assembly mass of 36.3 grams remains compatible with small robotic lander platforms, and the design is optimized for metal additive manufacturing processes.

The design process demonstrated the critical importance of material selection in payload-constrained aerospace applications. The initial aluminum shaft configuration was structurally inadequate with a Factor of Safety of only 0.40, requiring a design revision to titanium alloy. This material change, combined with a modest increase in shaft diameter, provided the necessary structural performance while adding only 8 grams to the total assembly mass. The modular cutting tip design, using carbide steel inserts mounted to a titanium body, represents a practical compromise that limits expensive, difficult-to-machine carbide to only the cutting edges while maintaining wear resistance where needed.

4.1 FUTURE WORK AND POTENTIAL IMPROVEMENTS

While the design satisfies the established requirements, several areas warrant further investigation given additional time and resources:

Experimental validation and ongoing collaboration. I'm currently working with JPL to explore manufacturing the drill using metal additive processes. If we can get it fabricated, I've developed a testing plan to validate the design under various soil conditions. The tests will evaluate how particle size (fine, medium, coarse), moisture content (25%, 50%, 75%), and pore size variation affect extraction efficiency and drilling performance. My test setup includes a benchtop drilling apparatus with a heating and condensing system to simulate the WISTLS instrument's water extraction process. I'll start with a calibration test using a controlled 75% humidity sample, heating it until no additional water is collected to establish a baseline extraction time. This will let me use a standardized time interval for all subsequent tests. The nine-condition test matrix will measure actual cutting torque during penetration, verify auger uptake efficiency, and assess whether sample retention is adequate or needs design modifications. I'll monitor water collection at defined intervals using a scale to characterize extraction behavior and identify when it levels off. This experimental work would confirm the analytical predictions and help optimize the auger geometry for better collection efficiency.

Finite element analysis. My hand calculations evaluated torsional stress in the shaft but didn't capture stress concentrations at geometric transitions like the shaft-to-bit interface or motor coupling. FEA would identify potential failure points that aren't visible in simplified analytical models. It could also examine combined loading scenarios, bending moments from lateral forces

and axial thrust during penetration, to ensure the shaft remains adequate under realistic conditions.

Thermal analysis. Drilling generates frictional heat that could volatilize water ice or other compounds the WISTLS instrument is designed to detect. The planned heating tests will provide some insight into thermal behavior during extraction, but more detailed modeling during the actual drilling phase would show whether temperature rise during penetration threatens sample integrity. If needed, I could incorporate design modifications like intermittent drilling cycles or thermal barriers to minimize heat transfer to the collected material.

Manufacturing process development. While the design incorporates features compatible with metal additive manufacturing, actual fabrication will reveal practical challenges with support structure removal, surface finish, and dimensional tolerances achievable with selective laser melting or electron beam melting. The ongoing JPL collaboration may provide insight into these manufacturing considerations, and prototyping would show what adjustments are needed to ensure the as-built geometry matches the design specifications.

Alternative auger configurations. The current design uses a constant-pitch helical flight with 180° angular coverage. The planned testing across different particle sizes and moisture contents may reveal whether changes to pitch, flight thickness, or alternative geometries (like tapered augers or dual-helix configurations) would improve collection efficiency or reduce torque. Results from the nine-condition test matrix could guide optimization beyond the initial design.

Bit geometry optimization. The dual-lobe cutting bit was selected based on literature recommendations for penetrating compacted soils, but alternative geometries, tri-lobe designs,

different cutting angles, or serrated edges, might reduce cutting forces or improve penetration in cemented regolith layers. If testing reveals excessive torque or poor penetration, I could experimentally compare multiple bit designs to find the optimal configuration for lunar conditions.

The current design represents a viable, manufacturing-ready solution that addresses the core requirements for lunar regolith sampling. The analytical validation confirms structural adequacy, and the design balances competing constraints of mass, strength, and manufacturability. The ongoing JPL collaboration to potentially manufacture and test the drill is an exciting opportunity to validate the design through the testing protocol I've developed. This work supports the broader development of cost-effective exploration technologies and ISRU strategies for future lunar missions.

5 DIVISION OF RESPONSIBILITIES

As the sole contributor to this project, I was responsible for all aspects of the WISTLS Sampling Drill design, analysis, and documentation. The work included concept development, geometric design, structural validation, material selection, and technical reporting. Working independently required careful time management and integration of multiple engineering disciplines to meet project objectives within the academic timeline.

My primary responsibilities included:

Preliminary research and problem definition. This phase involved reviewing NASA technical publications on lunar regolith mechanical properties [1,2], identifying mission constraints established by JPL, and formulating design requirements. Literature review informed the selection of analytical models for torque estimation and structural analysis.

Geometric design and CAD development. The drill geometry was developed iteratively using CAD software. Design parameters including outer diameter, auger pitch, shaft dimensions, and cutting bit configuration were established through volumetric analysis and packaging studies to satisfy the 20-gram sample requirement within spatial constraints.

Structural analysis and material selection. Torsional stress calculations were performed using mechanics of materials principles to evaluate candidate shaft configurations. The polar moment of inertia was calculated for proposed geometries, and maximum shear stress was determined under the estimated operational torque. Material comparison between Aluminum 6061-T6 and Titanium Ti-6Al-4V led to the final design revision based on factor of safety criteria.

Documentation and technical communication. All report sections, including background research, design rationale, analytical validation, and conclusions, were prepared in accordance with academic and professional standards. Reference materials were identified, evaluated, and properly cited following ASME guidelines.

Potential expansion with a collaborative team structure:

If this project had been undertaken by a team of three students, the scope could have been substantially expanded to include experimental validation and more comprehensive analysis. A collaborative structure would have enabled parallel development of complementary work while maintaining technical cohesion across the project.

A design-focused team member could have concentrated on geometric optimization and manufacturing preparation. This would include developing detailed additive manufacturing process plans, generating support structure strategies, investigating material anisotropy effects in metal additive manufacturing processes, and creating comprehensive technical drawings for fabrication. Additional design iterations exploring alternative cutting bit geometries or auger configurations could have been systematically evaluated.

An analysis-focused team member could have performed finite element analysis to validate analytical stress predictions and investigate stress concentrations at geometric transitions. Computational modeling would allow examination of combined loading scenarios including bending, axial thrust, and torsion. Fatigue analysis considering cyclic loading during extended drilling operations could provide insight into long-term structural durability. Parametric studies

varying shaft diameter, material properties, or geometric features would offer optimization opportunities beyond the scope of hand calculations.

A testing and validation-focused team member could have designed and executed experimental studies using lunar regolith simulants. Development of a benchtop drilling apparatus would enable measurement of actual cutting torques under controlled conditions. Thermal analysis could evaluate temperature rise during drilling to verify that volatile compounds would not be compromised. Physical prototyping using additive manufacturing would provide practical insight into fabrication challenges and dimensional accuracy achievable with the selected processes.

Such a team structure would have facilitated a more thorough investigation of the design space while distributing workload across specialized domains. Regular coordination between team members would ensure that geometric design decisions remained compatible with structural requirements and manufacturing constraints. Ultimately, a collaborative approach would have yielded a more comprehensively validated design with experimental data supporting analytical predictions.

REFERENCES

- [1] Williams, R. J., and Jadwick, J. J., 1980, Handbook of Lunar Materials, NASA Reference Publication 1057, NASA Lyndon B. Johnson Space Center, Houston, TX.
- [2] Carrier, W. D., Olhoeft, G. R., and Mendell, W., 1991, "Physical Properties of the Lunar Surface," *Lunar Sourcebook*, G. H. Heiken, D. T. Vaniman, and B. M. French, eds., Cambridge University Press, Cambridge, pp. 475–594.
- [3] Gere, J. M., and Goodno, B. J., 2013, *Mechanics of Materials*, 8th ed., Cengage Learning, Stamford, CT.
- [4] Budynas, R. G., and Nisbett, J. K., 2020, *Shigley's Mechanical Engineering Design*, 12th ed., McGraw-Hill Education, New York, NY.
- [5] ASTM International, 2020, "Standard Test Methods for Tension Testing of Metallic Materials," ASTM E8/E8M-20.
- [6] Aluminum Association, 2015, *Aluminum Standards and Data 2015*, Aluminum Association, Arlington, VA.
- [7] ASM International, 1990, *Titanium and Titanium Alloys, ASM Handbook*, Vol. 2, ASM International, Materials Park, OH.
- [8] Hajjar, J. F., 1998, "Torsional Behavior of Solid and Hollow Circular Shafts," *J. Struct. Eng.*, 124(2), pp. 161–170.
- [9] Iverson, R. M., 1997, "The Physics of Granular Flow," *Rev. Geophys.*, 35(4), pp. 245–296.
- [10] Heiken, G., Vaniman, D., and French, B. M., 1991, *Lunar Sourcebook: A User's Guide to the Moon*, Cambridge University Press, Cambridge.

APPENDIX A

This appendix provides the detailed volumetric analysis referenced in Section 2.1 that verifies the drill design can collect at least 20 grams of lunar regolith material.

A.1 DESIGN PARAMETERS

The auger geometry is defined by the following dimensions from Table 1:

- Sheath inner diameter: $D_{\text{sheath}} = 21.0 \text{ mm}$ (radius $r_{\text{sheath}} = 10.5 \text{ mm}$)
- Auger tube outer diameter: $D_{\text{tube}} = 7.5 \text{ mm}$ (radius $r_{\text{tube}} = 3.75 \text{ mm}$)
- Effective sampling length: $L = 100 \text{ mm}$
- Auger fin angular coverage: $\theta = 180^\circ$ (two 90° sections)

A.2 THEORETICAL VOLUME CALCULATION

The theoretical volume represents the maximum geometric storage capacity within the sheath, the raw annular volume between the inner wall of the sheath and the outer surface of the auger tube, before accounting for any internal components or fins that block space.

Cross-sectional area of the annular region:

$$A_{\text{annular}} = \pi(r_{\text{sheath}}^2 - r_{\text{tube}}^2)$$

$$A_{\text{annular}} = \pi(10.5^2 - 3.75^2) = \pi(110.25 - 14.0625) = \pi(96.1875)$$

$$A_{\text{annular}} = 302.2 \text{ mm}^2$$

Total annular volume over the 100 mm sampling length:

$$V_{\text{annular}} = A_{\text{annular}} \times L = 302.2 \times 100 = 30,220 \text{ mm}^3$$

Converting to cm^3 :

$$V_{\text{theoretical}} = \frac{30,220}{1000} = 30.2 \text{ cm}^3$$

This value represents the upper bound for volumetric capacity, the maximum possible sample volume if the entire annular space were available for regolith storage.

A.3 EFFECTIVE VOLUME ACCOUNTING FOR AUGER FINS

The auger fins occupy 180° of angular coverage (two 90° sections), which means they block approximately 50% of the annular cross-sectional area. The remaining 50% is available for sample collection:

$$V_{\text{effective}} = 0.5 \times V_{\text{theoretical}} = 0.5 \times 30.2 = 15.1 \text{ cm}^3$$

This calculation assumes:

- The helical fins occupy half the annular space
- Regolith fills the open regions with reasonable packing efficiency
- Clearance gaps (0.25 mm between fin and sheath) contribute negligibly to overall volume

A.4 ESTIMATED SAMPLE MASS

Using lunar regolith bulk density of $\rho = 1.5 \text{ g/cm}^3$ [1]:

$$m_{\text{sample}} = \rho \times V_{\text{effective}} = 1.5 \times 15.1 = 22.7 \text{ g}$$

A.5 MARGIN ABOVE REQUIREMENT

$$\text{Margin} = \frac{m_{\text{sample}} - m_{\text{required}}}{m_{\text{required}}} \times 100\% = \frac{22.7 - 20}{20} \times 100\% = 13.5\%$$

This margin accounts for uncertainties in packing efficiency, material loss during extraction, and variations in regolith density with depth. The design provides adequate capacity to ensure the 20 gram minimum requirement is met under realistic operating conditions.

A.6 SUMMARY

- **Theoretical Volume (30.2 cm³):** Maximum geometric capacity of the annular space between sheath and auger tube, assuming no internal obstructions
- **Effective Volume (15.1 cm³):** Usable sample capacity after accounting for 180° auger fin coverage blocking 50% of the space
- **Sample Mass (22.7 g):** Estimated collected mass using lunar regolith bulk density
- **Margin (13.5%):** Safety factor above the 20 g minimum requirement